

Exploring Spatial and Spatio-Temporal V-JEPA

2.1 Features for Aerial Depth Estimation and Semantic Scene Completion

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Abstract. Aerial 3D scene understanding is central to UAV navigation, mapping, inspection, search-and-rescue, human collaboration, and environmental monitoring. Camera-based perception is attractive for these settings because RGB cameras are lightweight, low-cost, widely available, and compatible with UAV size, weight, and power constraints. However, most visual 3D perception research remains centered on indoor or ground-level autonomous-driving scenes, while aerial imagery introduces altitude-dependent scale variation, oblique viewpoints, weak texture, and different occlusion patterns. Using OccuFly, a real-world aerial dataset with RGB images, metric depth maps, and semantic occupancy grids, we study whether V-JEPA 2.1 foundation features transfer to UAV 3D geometric and contextual perception. We compare spatial RGB and spatio-temporal video features for monocular depth estimation and depth-aware semantic scene completion. Our ablations progress from linear probing to DPT-style multiscale decoding, Simple Feature Pyramid adaptation, FiLM-based altitude and intrinsics conditioning, LSS-style 2D-3D lifting, and FoundationSSC/CGFormer-inspired occupancy variants. We further study pose-aware multi-view depth cost volumes constructed from calibrated neighboring frames. Through this staged analysis, we separate representation transfer, geometric conditioning, and 2D-3D lifting, deriving practical design lessons for aerial depth and semantic occupancy prediction.

Keywords: Aerial · 3D scene understanding · UAV perception · monocular depth estimation · semantic scene completion · semantic occupancy prediction · V-JEPA 2.1 · video foundation models · depth-aware 2D-3D lifting · pose-aware multi-view depth · OccuFly

1 Introduction

Aerial three-dimensional scene understanding is a central capability for unmanned aerial vehicles (UAVs) operating in navigation, mapping, infrastructure inspection, search-and-rescue, human collaboration, and environmental monitoring. In these settings, an autonomous platform must not only recognize visible image regions, but also infer metric geometry, free space, and the semantic occupancy of partially observed volumes. Such a representation is important for

collision avoidance, viewpoint planning, landing-zone assessment, and interaction with objects and humans in cluttered outdoor environments. Camera-based perception is particularly attractive for UAVs because RGB cameras are lightweight, inexpensive, widely available, and compatible with strict size, weight, and power constraints. However, inferring dense 3D structure from monocular or short video observations remains substantially harder than recognition, since the image formation process collapses scale, depth, and occlusion into a two-dimensional signal. Aerial video offers an additional source of geometry through motion parallax, but exploiting it requires more than generic temporal feature aggregation: the system must relate target and source views through camera calibration, relative pose, and depth hypotheses.

Most progress in semantic scene completion (SSC) and camera-based 3D perception has been driven by indoor reconstruction and autonomous-driving benchmarks. Early SSC formulations used depth as input to jointly predict volumetric occupancy and semantics [24]. Later monocular methods, such as MonoScene [5], moved the problem toward RGB-only 2D-to-3D lifting, while VoxFormer [12], TPVFormer [10], SurroundOcc [26], OpenOccupancy [25], CGFormer [31], and FoundationSSC [6] developed increasingly capable view-transformation, voxel-attention, and semantic-geometric fusion designs. These works demonstrate that explicit 3D representations are valuable for scene-level reasoning, but their primary evaluation domains remain road scenes with ground-level perspective, relatively constrained camera height, and characteristic automotive object layouts. Aerial viewpoints violate many of these assumptions: altitude changes the apparent scale of all objects, oblique views produce mixed top-down and side-view evidence, fine semantic boundaries can be weakly textured, and occlusions differ from those observed by ground vehicles. Consequently, strong methods trained or designed for terrestrial scenarios do not necessarily transfer to UAV perception without careful adaptation.

OccuFly [8] creates an opportunity to study this gap in a real-world aerial setting. It provides RGB images, metric depth maps, camera calibration, and semantic occupancy grids for aerial scenes captured at multiple altitudes and across urban, industrial, and rural environments. Unlike earlier UAV semantic segmentation datasets such as UAVid [15], which focus on 2D dense labeling, or aerial reconstruction resources such as UrbanScene3D [14], OccuFly directly supports the joint study of metric depth estimation and 3D semantic scene completion from aerial camera observations. This makes it possible to ask whether modern visual foundation representations provide transferable geometric and semantic priors for aerial 3D scene understanding, rather than only for image classification, 2D segmentation, or ground-level depth prediction. It also enables a more precise question: when multiple calibrated UAV views are available, does explicit multiview correspondence provide a stronger geometric prior for SSC than a monocular depth estimator alone?

In this work we investigate V-JEPA 2.1 [16] as a frozen foundation representation for aerial depth estimation and depth-aware semantic scene completion. V-JEPA and its successors learn by predicting representations in latent space

rather than reconstructing pixels [2, 3]. V-JEPA 2.1 further emphasizes dense predictive supervision, deep self-supervision at intermediate layers, and unified image/video tokenization, making it a natural candidate for dense 3D perception. The central question of this paper is not whether a large foundation model performs well on generic vision tasks, but whether its spatial and spatio-temporal features encode the aerial geometry needed for metric depth and volumetric semantic completion. We therefore keep the backbone frozen and vary the decoder, metadata conditioning, temporal aggregation, and geometric lifting mechanism.

We study this question along three axes. First, for monocular depth estimation, we compare frozen linear probes against structured DPT-style decoders [21]. The linear probes test how linearly accessible metric depth is from V-JEPA 2.1 features, while the DPT decoders test whether multiscale token reassembly and convolutional fusion can recover fine spatial detail from frozen transformer features. We also examine a Simple Feature Pyramid (SFP) adapter inspired by ViTDet [11], which converts a single final-layer feature map into a pyramid suitable for dense prediction. Because aerial scale is strongly coupled to flight altitude and camera calibration, we further condition selected decoders with feature-wise linear modulation (FiLM) [18] using altitude and camera intrinsics. Second, we compare spatial RGB features with spatio-temporal video features. Aerial videos contain motion parallax and temporal continuity that may help resolve depth ambiguities, but such cues are not guaranteed to be preserved in frozen video representations. We therefore contrast image-only features, repeated-frame controls, shuffled temporal controls, spatio-temporal linear probes, and dense temporal attention adapters.

Third, for SSC, we compare monocular and multiview geometric priors inside depth-aware lifting architectures inspired by Lift-Splat-Shoot [19], CGFormer [31], and FoundationSSC [6]. The monocular V-JEPA/FoundationSSC branch uses a DPT-FiLM depth-probability frontend to lift semantic context into the OccuFly voxel grid. We further introduce a Pose-MVS-Cost-Volume FoundationSSC variant that replaces the monocular depth-probability frontend with a posed multiview stereo cost volume in the spirit of learned MVS cost-volume methods [29]. Given a target UAV view and neighboring calibrated source views, this branch matches target/source V-JEPA video features across metric depth hypotheses, regularizes the resulting cost volume, and converts it into the depth distribution consumed by the FoundationSSC lifting module. We also evaluate a Pose-MVS-FiLM variant that applies altitude/intrinsics FiLM conditioning directly to the MVS depth-bin logits before the depth softmax, so that the dense depth map and lifting probabilities remain consistent.

This design separates two different uses of video. Dense temporal attention tests whether frozen V-JEPA video tokens contain useful target-frame depth cues after nonlinear aggregation. In contrast, Pose-MVS tests an explicitly geometric hypothesis: calibrated source views should constrain depth through projective correspondence, even when monocular appearance cues are ambiguous. This distinction is important for aerial perception, where altitude variation, repeated

textures, vegetation, roofs, and oblique viewing directions can make single-image depth uncertain while neighboring UAV frames may provide exploitable parallax.

Our contributions are as follows:

- We present a systematic study of frozen V-JEPA 2.1 spatial and spatio-temporal representations for aerial metric depth estimation on OccuFly.
- We compare representation probes of increasing structure: linear depth heads, DPT-style multiscale decoders, SFP-adapted final-layer decoders, and altitude/intensity-conditioned FiLM variants.
- We analyze the role of temporal evidence through video-feature probes and temporal-attention adapters, including controls that separate true temporal context from repeated or shuffled frame inputs.
- We extend the investigation from depth to semantic scene completion by incorporating predicted depth and depth distributions as geometric priors in LSS-, CGFormer-, and FoundationSSC-inspired aerial SSC variants.
- We introduce Pose-MVS-Cost-Volume FoundationSSC, which replaces monocular depth-probability lifting with pose-aware multiview cost-volume depth estimation from target/source UAV views.
- We further evaluate Pose-MVS-FiLM, a metadata-conditioned MVS variant that modulates cost-volume depth logits with altitude and camera intrinsics while preserving probability–depth consistency for voxel lifting.
- We identify the practical domain gap between ground-level camera-centric 3D perception and aerial 3D scene understanding, providing a focused evaluation protocol for future foundation-model transfer studies in UAV perception.

2 Related Work

2.1 Aerial scene understanding and aerial 3D benchmarks

Aerial perception has traditionally been studied through 2D semantic segmentation, object detection, mapping, and reconstruction. UAVid [15] introduced high-resolution oblique UAV videos for semantic segmentation and highlighted challenges such as scale variation, moving objects, and temporal consistency. Other aerial resources, including UrbanScene3D [14], support large-scale urban reconstruction and simulation from aerial observations. These datasets are valuable for 2D and reconstruction-oriented aerial perception, but they do not directly provide the combination of RGB images, metric depth, camera calibration, and per-frame semantic occupancy grids needed to study aerial SSC.

OccuFly [8] addresses this limitation by introducing a real-world aerial benchmark for camera-based 3D semantic scene completion. It provides aerial RGB images, metric depth maps, semantic voxel grids, and camera metadata over multiple scenes, altitudes, environments, and seasons. The dataset is especially relevant for UAV perception because it captures the geometric ambiguity of low-altitude flight: object scale depends on altitude, many surfaces are seen from oblique or partially top-down viewpoints, and free-space reasoning must

be performed in a 3D volume rather than only on the image plane. Our work uses OccuFly not merely as an evaluation dataset, but as a testbed for studying whether dense image/video foundation features and explicit multiview geometry transfer to aerial depth and semantic occupancy.

2.2 Monocular depth estimation and dense prediction decoders

Monocular depth estimation is inherently ambiguous because metric depth must be inferred from perspective, texture, object size, scene layout, and learned priors. Early deep approaches used coarse-to-fine convolutional networks and scale-aware objectives [7]. Later methods improved dense prediction through stronger encoders, multiscale decoders, and more diverse training data. MiDaS demonstrated that mixing heterogeneous datasets with compatible objectives improves cross-dataset robustness [22]. DPT replaced purely convolutional backbones with vision transformers and reassembled intermediate tokens into multiscale image-like features for dense prediction [21]. ZoeDepth combined relative-depth generalization with metric-depth specialization [4], while Depth Anything and Depth Anything V2 scaled monocular depth training with large pseudo-labeled and synthetic data, producing robust general-purpose depth models [27, 28].

These models show that large-scale pretraining and dense decoders are crucial for depth, but aerial depth remains a distinct transfer problem. UAV images often lack the ground-level cues used by indoor and automotive models: the horizon may be absent, object size priors vary with altitude, repetitive roofs and vegetation create weak texture, and far-range depth errors can dominate voxel lifting. For this reason, our work does not treat a single depth model as a black-box solution. Instead, we compare lightweight linear probes with DPT-style decoders on frozen V-JEPA 2.1 features, evaluate final-layer SFP adaptation inspired by ViTDet [11], and condition selected decoders with altitude and camera intrinsics through FiLM [18]. This lets us separate three factors: the geometric information already present in the representation, the spatial reconstruction capacity of the decoder, and the usefulness of explicit aerial metadata.

2.3 Multiview stereo and cost-volume geometry

Monocular depth relies heavily on learned priors, whereas multiview stereo (MVS) estimates depth by matching observations across calibrated views. Learning-based MVS methods such as MVSNet build a plane-sweep cost volume by warping source-view features into a reference view over multiple depth hypotheses, regularize the cost volume with 3D convolutions, and infer depth from a probability distribution over planes [29]. Recurrent MVSNet reduces the memory footprint of high-resolution MVS by replacing full 3D cost-volume regularization with recurrent processing over depth hypotheses [30]. These methods are designed for depth reconstruction, but the underlying principle is directly relevant to aerial SSC: when camera poses are available, target/source correspondence can provide geometric evidence that is complementary to monocular appearance priors.

Our Pose-MVS variants adapt this cost-volume principle to frozen V-JEPA video features and downstream semantic scene completion. Rather than producing only a target-view depth map, the MVS branch produces a categorical depth distribution aligned with the FoundationSSC lifting bins. The resulting distribution is used to place semantic context into the OccuFly voxel grid, making MVS a geometric frontend for SSC rather than a standalone reconstruction module. This differs from the dense temporal attention depth experiments: temporal attention learns how to aggregate video tokens, while Pose-MVS explicitly projects source views into the target camera using intrinsics, relative pose, and metric depth hypotheses. The Pose-MVS-FiLM variant further asks whether altitude and camera calibration, already useful for monocular DPT-FiLM depth, should also modulate multiview cost-volume logits before depth probabilities are formed.

2.4 Self-supervised visual foundation models

Self-supervised visual foundation models aim to learn transferable representations without task-specific labels. Masked autoencoders showed that high-ratio masked image reconstruction can scale vision transformers effectively [9]. I-JEPA introduced a non-generative joint-embedding predictive architecture for images, predicting target-block representations from context representations instead of reconstructing pixels [1]. DINOv2 demonstrated that carefully curated large-scale self-supervised training can yield robust all-purpose image features [17], and DINOv3 further emphasizes high-resolution dense features and broad transfer across natural and aerial imagery [23].

Video representation learning extends these ideas to temporal structure. V-JEPA learns video representations by predicting masked spatio-temporal features in latent space and can be used as a frozen backbone with lightweight probes [3]. V-JEPA 2 scales this approach to internet-scale video and image data for understanding, prediction, and planning [2]. V-JEPA 2.1 further modifies the recipe toward dense image/video features through dense predictive loss, deep self-supervision, and multimodal tokenizers [16]. These properties make V-JEPA 2.1 attractive for aerial 3D perception, where one needs spatial layout, temporal consistency, and view-dependent geometry. However, strong generic dense features do not guarantee metric depth accuracy or reliable 3D lifting under aerial viewpoints. Our work therefore evaluates V-JEPA 2.1 through explicit downstream depth, cost-volume, and SSC probes rather than only through representation visualization or global recognition metrics.

2.5 2D-to-3D lifting and camera-based semantic scene completion

Semantic scene completion jointly estimates occupied and free voxels together with semantic labels. SSCNet introduced the task using single-view depth input and 3D convolutional reasoning [24]. MonoScene extended SSC to monocular RGB by projecting 2D features into 3D and using 3D context priors [5]. Camera-based outdoor SSC then evolved toward sparse voxel queries, deformable

attention, and explicit view transformation. VoxFormer uses estimated depth to initialize sparse visible voxel queries and then densifies them with transformer reasoning [12]. TPVFormer represents 3D space through three orthogonal planes, improving over a single BEV plane for semantic occupancy prediction [10]. SurroundOcc and OpenOccupancy expanded the problem toward surrounding multi-camera occupancy perception in autonomous driving [25, 26].

A recurring theme in these works is the importance of depth-aware lifting. Lift-Splat-Shoot first formalized a camera-to-BEV representation by predicting a depth distribution for each image feature and splatting frustum features into a common spatial grid [19]. BEVDepth later showed that explicit depth supervision improves camera-based BEV 3D detection [13]. For SSC, CGFormer addresses depth ambiguity by generating context-dependent voxel queries and extending deformable cross-attention from 2D image space to 3D pixel space; it also combines voxel and tri-perspective-view representations to capture local and global 3D context [31]. FoundationSSC argues that semantic and geometric priors should be decoupled at both source and pathway levels, using foundation features and geometric depth distributions before hybrid 3D fusion [6].

Our SSC design follows this line of work but differs in domain and experimental goal. Existing methods are primarily optimized for indoor or autonomous-driving scenes, whereas we target aerial observations from OccuFly. We use predicted monocular depth and depth probabilities as geometric priors for voxel lifting, while V-JEPA 2.1 features provide semantic context. We then add Pose-MVS-Cost-Volume FoundationSSC to test whether calibrated target/source matching improves the depth distribution used for lifting, and Pose-MVS-FiLM to test whether altitude and intrinsics should condition that multiview distribution. This allows us to compare three sources of geometry under a shared aerial SSC protocol: monocular depth priors, video-feature temporal aggregation, and explicit posed multiview correspondence.

2.6 Position of this work

The closest work to ours is OccuFly [8], which establishes the aerial SSC benchmark and reports baseline limitations of existing methods. Our work is complementary: rather than proposing a new dataset, we use OccuFly to analyze the transfer behavior of a recent video/image foundation model for coupled aerial 3D tasks. Compared with generic monocular depth models such as Depth Anything V2 [28], we focus on what can be recovered from frozen V-JEPA 2.1 features and how decoder structure, metadata conditioning, temporal context, and posed multiview matching affect metric depth. Compared with camera-based SSC methods such as MonoScene, VoxFormer, CGFormer, and FoundationSSC [5, 6, 12, 31], we explicitly study aerial depth priors and foundation features under UAV-specific viewpoint and scale shifts. Compared with standard MVS methods such as MVSNet and R-MVSNet [29, 30], our goal is not standalone depth reconstruction but semantic occupancy: the cost volume is used to produce the depth-probability tensor that drives FoundationSSC-style voxel lifting.

The resulting study connects four previously separate threads—aerial benchmarks, dense foundation representations, multiview cost-volume geometry, and depth-aware semantic scene completion—into a single evaluation of aerial 3D scene understanding.

3 Method

This section describes our evaluation framework for transferring frozen V-JEPA 2.1 features to aerial metric depth estimation and depth-aware semantic scene completion (SSC). The method is intentionally organized as a sequence of probes and progressively more structured decoders. This design separates four factors that are often confounded in end-to-end training: (i) the amount of aerial geometry already encoded in the frozen foundation representation, (ii) the decoder structure required to recover dense metric predictions from transformer tokens, (iii) the usefulness of predicted depth as a geometric prior for volumetric semantic occupancy, and (iv) whether explicit multiview correspondence improves the 3D prior beyond monocular depth estimation, and (v) whether altitude/intrinsics FiLM conditioning should also modulate the multiview geometry prior. We use OccuFly [8] throughout because it provides RGB images, metric depth maps, camera calibration, semantic occupancy grids, and, for the pose-aware MVS branch, per-frame camera poses from real UAV viewpoints.

3.1 Problem Formulation and Notation

An OccuFly sample is represented as

$$s_t = (I_t, D_t, M_t, Y_t, K_t, h_t, T_t^{w \rightarrow c}), \quad (1)$$

where $I_t \in \mathbb{R}^{3 \times H \times W}$ is an RGB frame, D_t is a metric depth map, $M_t \in \{0, 1\}^{H \times W}$ is the valid-depth mask, Y_t is the semantic occupancy grid, K_t is the camera intrinsic matrix, h_t is the UAV altitude, and $T_t^{w \rightarrow c} \in \mathbb{R}^{4 \times 4}$ is the world-to-camera pose used by the multiview-stereo branch when pose supervision is available. All depth experiments resize images to 384×384 , which gives a 24×24 patch grid for a ViT backbone with patch size 16. Depth values are clipped to $[d_{\min}, d_{\max}] = [10^{-3}, 80]$ m for metric depth training and evaluation.

Depth supervision is resized with mask-aware interpolation rather than ordinary interpolation. Given a depth map D and validity mask M , the resized depth and mask are

$$D' = \frac{\text{resize}(D \odot M)}{\max(\text{resize}(M), \epsilon)}, \quad M' = \mathbf{1}\{\text{resize}(M) > 0\}, \quad (2)$$

where losses and metrics are computed only on $\Omega_D = \{(u, v) : M'_{u,v} = 1\}$. In the implementation, pixels outside the valid resized mask are assigned zero depth and ignored by all depth losses.

For SSC, OccuFly occupancy grids are stored in a raw width-height-depth layout and converted to the internal depth-height-width layout used by our training code:

$$Y_{\text{WHD}} \in \{0, \dots, 36, 255\}^{192 \times 128 \times 128}, \quad Y_{\text{DHW}} \in \{0, \dots, 36, 255\}^{128 \times 128 \times 192}. \quad (3)$$

The voxel grid covers

$$x \in [-48, 48] \text{ m}, \quad y \in [-32, 32] \text{ m}, \quad z \in [0, 64] \text{ m}, \quad \Delta = 0.5 \text{ m}. \quad (4)$$

Label 0 denotes empty/free space, and label 255 denotes ignored or unknown voxels. We write the semantic output space as $\mathcal{C}$ because it depends on the active SSC configuration: raw-ID-compatible FoundationSSC runs use raw slots $0, \dots, 36$, while CGFormer-style reporting remaps raw IDs to a contiguous empty-plus-occupied taxonomy before computing semantic metrics. In all cases, 0 denotes empty/free space and $\mathcal{C}_{\text{occ}} = \mathcal{C} \setminus \{0\}$ denotes occupied semantic classes.

The monocular depth and SSC task functions are

$$f_D : (I_t, K_t, h_t) \mapsto \hat{D}_t, \quad f_S^{\text{mono}} : (I_t, K_t, h_t, \hat{D}_t, P_t^D) \mapsto \hat{Y}_t, \quad (5)$$

where P_t^D is a categorical or Gaussian depth probability tensor used for geometric lifting. For pose-aware multiview SSC, the target frame is paired with N_s posed source views

$$\mathcal{V}_t = \{(I_{t_i}, K_{t_i}, T_{t_i}^{w \rightarrow c})\}_{i=0}^{N_s}, \quad t_0 = t, \quad (6)$$

and the task function becomes

$$f_S^{\text{MVS}} : (\mathcal{V}_t, h_t) \mapsto \hat{Y}_t. \quad (7)$$

In the multiview branches, the depth prior is produced by a pose-aware cost volume rather than by a monocular DPT-FiLM depth head. The FiLM-conditioned variant further modulates the MVS depth-bin logits with altitude/intrinsics metadata before depth probabilities are formed. We keep the V-JEPA 2.1 backbone frozen in all experiments. This frozen-backbone protocol follows the common foundation-model transfer setting and makes the comparison diagnostic: improvements can be attributed to decoder design, metadata conditioning, temporal aggregation, depth-prior use, or multiview geometric matching rather than to representation finetuning.

3.2 Frozen V-JEPA 2.1 Feature Extraction

Let E_θ denote the frozen V-JEPA 2.1 encoder [16]. For an image input, the implementation uses the video-capable V-JEPA code path with a singleton temporal axis and extracts spatial patch maps. For the ViT-Gigantic checkpoint used in the main experiments, the channel dimension is $C = 1664$, the model

has 48 transformer blocks, and a 384×384 image produces a 24×24 patch map. We use either the final layer alone or four intermediate layers:

$$\mathcal{L}_G = \{11, 23, 37, 47\}, \quad F_t^{(l)} = E_\theta^{(l)}(I_t) \in \mathbb{R}^{C \times 24 \times 24}, \quad l \in \mathcal{L}_G. \quad (8)$$

For the 40-block giant checkpoint used by helper code, the same relative-depth intent is mapped to $\{9, 19, 29, 39\}$. This checkpoint-aware mapping avoids accidentally requesting nonexistent transformer layers while preserving the early/mid/late feature sampling pattern.

For video experiments, a clip centered at target frame t is

$$X_t = [I_{t+\tau_1}, \dots, I_{t+\tau_T}] \in \mathbb{R}^{3 \times T \times 384 \times 384}. \quad (9)$$

With V-JEPA tubelet size 2, the encoder returns

$$G_t^{(l)} = E_\theta^{(l)}(X_t) \in \mathbb{R}^{C \times T' \times 24 \times 24}, \quad T' = T/2. \quad (10)$$

For the implemented four-frame experiments, the target-centered real clip uses offsets $[-1, 0, 0, +1]$. Thus the target frame is included twice; the repeated-frame control uses four copies of the target frame, and the shuffled-frame control deterministically shuffles the same target-centered indices. We report these controls because they distinguish genuine temporal information from gains caused by repeated target evidence, batch-normalization effects, or increased input token count.

3.3 Depth Estimation Heads

All depth heads produce a dense metric depth prediction. Most variants first predict logits over $K_D = 256$ depth bins and then convert them to metric depth by expectation,

$$p_{k,u,v} = \text{softmax}_k(z_{k,u,v}), \quad \hat{D}_{u,v} = \sum_{k=1}^{K_D} p_{k,u,v} d_k, \quad (11)$$

where d_k are bin centers. The resulting map is resized to the supervision resolution and trained with a valid-pixel scale-invariant logarithmic loss. For $N = |\Omega_D|$ valid pixels and $g_i = \log \hat{D}_i - \log D_i$, the loss is

$$\mathcal{L}_{\text{sig}} = \sqrt{\frac{1}{N} \sum_{i \in \Omega_D} g_i^2 - \lambda_{\text{si}} \left(\frac{1}{N} \sum_{i \in \Omega_D} g_i \right)^2} + \epsilon, \quad (12)$$

following the scale-invariant depth objective used in monocular depth estimation [7]. We use this loss because aerial metric depth contains unavoidable global-scale ambiguity, while the altitude/intrinsics conditioning experiments test whether that ambiguity can be reduced by physically meaningful metadata.

Image Linear Probe The simplest probe tests whether metric depth is linearly accessible from the frozen final-layer feature map. Given $F_t^{(47)} \in \mathbb{R}^{1664 \times 24 \times 24}$, the head is a synchronized batch-normalization layer followed by a 1×1 convolution:

$$Z_t(:, u, v) = W \text{BN}(F_t^{(47)}(:, u, v)) + b, \quad W \in \mathbb{R}^{256 \times 1664}. \quad (13)$$

No spatial decoder, skip connection, or trainable backbone parameter is used. This probe is intentionally restrictive: if it succeeds, depth is already directly organized in the final V-JEPA feature space; if it fails relative to structured decoders, the failure indicates that aerial geometry is present only in a form requiring spatial reassembly or nonlinear decoding. The same linear-probe implementation also supports DINOv3-style dense prediction utilities, which provide the batch-normalized linear head and features-to-depth conversion used in our frozen-probe baseline [23].

Spatio-Temporal Linear Probe The spatio-temporal linear probe (ST-LP) is the video analogue of the image linear probe. It asks whether V-JEPA 2.1 video features expose temporal depth cues in a form that can be decoded by a single location-wise linear map. For a four-frame clip, $T' = 2$ after tubeletization, and the selected final-layer tensor is

$$G_t^{(47)} \in \mathbb{R}^{1664 \times 2 \times 24 \times 24}. \quad (14)$$

We concatenate the temporal tubelet axis into channels:

$$\tilde{F}_t \in \mathbb{R}^{(2 \cdot 1664) \times 24 \times 24}, \quad \tilde{F}_t(t'C + c, u, v) = G_t^{(47)}(c, t', u, v), \quad (15)$$

then apply the same batch-normalized 1×1 depth-bin head as in Eq. (13). The trainable head therefore has 3328 input channels for the four-frame setting. Because the operation is still linear in the frozen spatio-temporal features, ST-LP is a controlled diagnostic for temporal representation quality rather than a high-capacity video decoder.

DPT-FiLM Multiscale Depth Decoder The structured image decoder follows the Dense Prediction Transformer (DPT) principle of reassembling transformer tokens from multiple layers into image-like feature maps and fusing them with convolutional refinement [21]. For the four V-JEPA layers in Eq. (8), each feature map is reassembled and projected to a common decoder width $C_D = 256$:

$$R_l = \rho_l \left(F_t^{(l)} \right), \quad R_l \in \mathbb{R}^{C_D \times H_l \times W_l}, \quad l \in \mathcal{L}_G, \quad (16)$$

where ρ_l denotes the DPT reassembly and projection block. The multiscale maps are then modulated by altitude and intrinsics through FiLM [18].

The metadata vector is

$$m_t = \left[\frac{h_t - 40}{10}, \frac{f_x}{W}, \frac{f_y}{H}, \frac{c_x}{W} - 0.5, \frac{c_y}{H} - 0.5 \right] \in \mathbb{R}^5, \quad (17)$$

where (f_x, f_y, c_x, c_y) are camera intrinsics expressed relative to the image size used by the branch. A small MLP maps m_t to an embedding $e_t \in \mathbb{R}^{128}$. For each DPT level,

$$\tilde{R}_l = R_l \odot (1 + \gamma_l(e_t)) + \beta_l(e_t), \quad (18)$$

where γ_l and β_l are learned affine projections. We initialize the FiLM projections so that $\gamma_l \approx 0$ and $\beta_l \approx 0$ at the start of training, making the initial decoder identical to the unconditioned DPT path. This avoids early training instability and lets the model learn when altitude or calibration should alter the feature scale.

The FiLM-modulated maps are fused by DPT refinement blocks and decoded into depth logits. This decoder tests whether frozen V-JEPA layers contain spatially distributed aerial geometry that is not linearly accessible from the final map. It also tests a physically motivated hypothesis: because UAV altitude and camera intrinsics determine the mapping from pixels to metric rays, exposing them to the decoder should improve metric scale recovery under altitude-dependent appearance changes.

Final-Layer Simple Feature Pyramid DPT-FiLM The final-layer Simple Feature Pyramid (SFP) variant isolates whether the last V-JEPA layer alone is sufficient when equipped with an explicit multiscale adapter. Instead of extracting four transformer layers, we use only

$$F_t^{(47)} \in \mathbb{R}^{1664 \times 24 \times 24}. \quad (19)$$

A Simple Feature Pyramid inspired by ViTDet [11] converts this single map into four feature levels:

$$\text{SFP}(F_t^{(47)}) = \{S_4, S_2, S_1, S_{1/2}\}, \quad S_s \in \mathbb{R}^{256 \times (24s) \times (24s)}, \quad s \in \{4, 2, 1, 1/2\}. \quad (20)$$

For a 384×384 input, these levels have spatial sizes approximately 96×96 , 48×48 , 24×24 , and 12×12 . Unlike the four-layer DPT-FiLM decoder, this branch does not apply DPT reassembly to raw V-JEPA intermediate layers; the SFP itself creates the multiscale maps. The resulting maps are projected, FiLM-conditioned with Eq. (18), fused by the DPT depth head, and converted to metric depth. This experiment distinguishes two possible explanations for DPT gains: multi-layer semantic hierarchy versus final-layer features plus a learned multiscale pyramid.

Video DPT-FiLM with Dense Temporal Attention ST-LP is deliberately low capacity. To test whether temporal evidence becomes useful when aggregated nonlinearly, we also evaluate a video DPT-FiLM decoder with dense temporal attention (DTA). For each selected layer $l \in \mathcal{L}_G$, V-JEPA returns

$$G_t^{(l)} \in \mathbb{R}^{1664 \times T' \times 24 \times 24}. \quad (21)$$

At each spatial site (u, v) , the temporal sequence is

$$S_{t,u,v}^{(l)} = \left[G_t^{(l)}(:, 1, u, v), \dots, G_t^{(l)}(:, T', u, v) \right] \in \mathbb{R}^{T' \times 1664}. \quad (22)$$

The DTA adapter first reduces channels with a $1 \times 1 \times 1$ projection to an adapter width $A = 512$, applies LayerNorm, and uses a learned query to attend over the temporal tokens with multi-head attention. The attended token is refined by an MLP residual, expanded back to 1664 channels, and interpolated with the temporal mean:

$$\bar{F}_t^{(l)}(:, u, v) = \mu_t^{(l)}(:, u, v) + \alpha \left(\text{Expand}(\text{MLP}(\text{MHA}(q, K, V))) - \mu_t^{(l)}(:, u, v) \right), \quad (23)$$

where $\mu_t^{(l)} = \text{mean}_\tau G_t^{(l)}(:, \tau, :, :)$ and $\alpha = \sigma(a)$ is initialized to approximately 0.1. The output maps $\{\bar{F}_t^{(l)}\}_{l \in \mathcal{L}_G}$ are then passed to the same DPT-FiLM decoder as the image-mode multiscale model. The residual gate makes the initial model close to temporal averaging, while allowing training to learn stronger temporal aggregation if motion parallax or temporal continuity is useful.

3.4 Depth Experiment Design

Table 1 summarizes the depth variants and the question each variant is designed to answer. The experiments are not merely architectural alternatives; they form a controlled sequence from minimal representation probing to structured spatio-temporal decoding.

The standard depth split uses scenes 01–05 for training, scenes 06–07 for validation, and scenes 08–09 for testing, with altitudes 30 m, 40 m, and 50 m. We also include an altitude-interpolation protocol in which training uses 30 m and 50 m samples from scenes 01–08, validation uses 40 m samples from scenes 01–04, and testing uses 40 m samples from scenes 05–08:

$$\mathcal{D}_{\text{train}} = \{s_t : h_t \in \{30, 50\}, \text{scene}(t) \in \{01, \dots, 08\}\}, \quad (24)$$

$$\mathcal{D}_{\text{val}} \subset \{s_t : h_t = 40, \text{scene}(t) \in \{01, 02, 03, 04\}\}, \quad (25)$$

$$\mathcal{D}_{\text{test}} \subset \{s_t : h_t = 40, \text{scene}(t) \in \{05, 06, 07, 08\}\}. \quad (26)$$

This split is a scale-generalization stress test: the model must interpolate to the middle altitude rather than merely memorizing a random mixture of all flight heights.

3.5 Depth Probabilities for Semantic Scene Completion

The SSC models require not only a dense depth expectation $\hat{D}$ but also a probability distribution over depth bins for frustum lifting. Let $\mathcal{B} = \{d_1, \dots, d_K\}$ denote the bin centers expected by the lifting module. We use four probability sources depending on the branch. When a monocular depth head directly outputs

Table 1: Depth-estimation variants and the transfer question isolated by each design.

Variant	Trainable components	Purpose
Image probe	linear BN and 1×1 depth-bin head on final V-JEPA features	Tests whether metric aerial depth is linearly accessible from the frozen final spatial representation.
ST-LP probe	video BN and 1×1 head after temporal-channel concatenation	Tests whether frozen video features expose temporal depth cues without a nonlinear video decoder.
DPT-FiLM	DPT reassembly/fusion, depth head, metadata assembly, FiLM layers	Tests whether multiscale token reassembly and altitude/intrinsics conditioning recover dense metric structure.
Final-layer DPT-FiLM	SFP SFP adapter, DPT-style fusion, FiLM, depth head	Tests whether a final-layer single-scale V-JEPA map can be converted into an effective dense pyramid.
Video FiLM+DTA	DPT-DTA adapters, DPT-FiLM decoder, depth head	Tests whether nonlinear temporal aggregation improves depth beyond image-mode decoding and linear temporal probing.

bin logits, we use the softmax probabilities from Eq. (11). When the FoundationSSC LSS module expects a different bin definition, we rebuild a Gaussian distribution around the dense depth prediction:

$$P_{k,u,v}^D = \frac{\exp\left(-\frac{(d_k - \hat{D}_{u,v})^2}{2\sigma_D^2}\right)}{\sum_{j=1}^K \exp\left(-\frac{(d_j - \hat{D}_{u,v})^2}{2\sigma_D^2}\right)}. \quad (27)$$

For the FoundationSSC hybrid, the LSS depth centers are defined by

$$\text{dbound} = [0, 64, 0.25], \quad (28)$$

which gives 256 depth bins over the 64 m OccuFly lifting range. If the frontend and LSS bin counts differ, the implementation resizes the probability tensor along the depth dimension and renormalizes it. If the bin count already matches, the Gaussian reconstruction in Eq. (27) preserves the dense-depth expectation to numerical precision. This remapping is important because a small inconsistency between the depth distribution and the expected metric depth can translate into systematic voxel misplacement during lifting. The unconditioned pose-MVS branch bypasses this Gaussian reconstruction: it constructs $P_t^{D,MVS}$ directly by softmax-normalizing a regularized projective cost volume over metric depth-bin centers, and obtains $\hat{D}_t^{MVS}$ by expectation. The Pose-MVS-FiLM branch applies FiLM to the MVS depth-bin logits before the softmax and then computes dense depth from the resulting conditioned probability distribution. Thus the LSS

probability volume and the dense depth map used by voxel proposal describe the same ray-wise depth distribution.

3.6 Depth-Aware Semantic Scene Completion

Our SSC study evaluates how depth priors can be used as geometric evidence for aerial semantic occupancy. We implement four complementary branches. The first is a CGFormer-based depth-prior comparison, which isolates the value of different monocular depth priors under a fixed CGFormer-style evaluation pipeline [31]. The second is a monocular V-JEPA/F2D/FoundationSSC hybrid that performs in-model probability lifting with real FoundationSSC modules [6]. The third is a pose-aware multiview-stereo FoundationSSC hybrid that replaces the monocular depth-probability frontend with an explicit projective cost-volume estimator in the style of learned MVS cost volumes [29]. The fourth, Pose-MVS-FiLM, keeps the same projective MVS geometry but FiLM-conditions the MVS depth-bin logits using altitude and camera intrinsics. The distinction matters: the CGFormer branch compares external or precomputed depth priors, the monocular FoundationSSC branch tests learned single-image depth uncertainty, the pose-MVS branch tests calibrated multiview correspondence, and the Pose-MVS-FiLM branch tests whether metadata conditioning improves that multiview prior.

CGFormer Depth-Prior Comparison CGFormer addresses camera-based SSC by combining context-aware voxel queries and geometry-aware transformer reasoning [31]. In our implementation, we use the CGFormer code path as a controlled SSC comparison layer with OccuFly preparation, depth-prior export, checkpoint evaluation, and pairwise comparison utilities. The depth-prior variants include a Depth Anything V2 prior [28] and V-JEPA 2.1 DPT-FiLM priors exported from our altitude-conditioned depth estimator. Holding the SSC evaluation path fixed while changing the prior allows us to ask whether V-JEPA-derived depth is more useful for aerial occupancy than a strong generic depth foundation model.

For CGFormer reporting, raw OccuFly IDs are remapped to a contiguous 22-label taxonomy consisting of empty/free space plus 21 occupied semantic classes. Label 255 is ignored. Completion metrics are computed on the binary occupied-vs-empty task, while semantic mIoU is computed only over occupied classes. The currently implemented masked 3D projector variant uses independent random voxel masking; we therefore describe it as such and do not claim a mixed block/column/frustum masking strategy.

V-JEPA/F2D/FoundationSSC Hybrid The integrated SSC model combines a frozen V-JEPA backbone, a features-to-depth (F2D) DPT-FiLM frontend, a geometry-aware 2D context adapter, and real FoundationSSC lifting and

3D decoding modules. Its forward pass is

$$(I_t, m_t, K_t) \rightarrow \{F_t^{(l)}\}_{l \in \mathcal{L}_G} \rightarrow (\hat{D}_t, P_t^D, F_t^{\text{DPT}}) \rightarrow C_t \rightarrow (V_t^{\text{LSS}}, V_t^{\text{Vox}}) \rightarrow \hat{Y}_t. \quad (29)$$

Here F_t^{DPT} denotes the projected DPT feature produced by the depth frontend, and $C_t \in \mathbb{R}^{128 \times 48 \times 48}$ is the fused 2D context map used for lifting.

The context adapter projects each selected V-JEPA level to a common 48×48 grid and fuses the projected features with the DPT feature:

$$C_t = \phi_{\text{ctx}} \left(\left[\text{up}_{48} \left(\pi_l(F_t^{(l)}) \right) \right]_{l \in \mathcal{L}_G} \oplus \text{up}_{48} \left(\pi_D(F_t^{\text{DPT}}) \right) \right), \quad (30)$$

where π_l and π_D are learned projections, up_{48} denotes resizing to the 48×48 feature grid, $\oplus$ is channel concatenation, and ϕ_{ctx} is the fusion network. This design follows the source/pathway separation advocated by FoundationSSC [6]: V-JEPA features provide semantic context, while P_t^D supplies explicit geometric uncertainty.

LSS-style frustum lifting. Lift-Splat-Shoot (LSS) constructs frustum features by combining a context vector with a categorical depth distribution at each image location [20]. For context $C_t \in \mathbb{R}^{C_c \times H_f \times W_f}$ and depth probabilities $P_t^D \in \mathbb{R}^{K \times H_f \times W_f}$, the lifted frustum tensor is

$$\Phi_{c,k,u,v} = C_t(c, u, v) P_t^D(k, u, v). \quad (31)$$

Using resized intrinsics (f_x, f_y, c_x, c_y) and feature-cell centers

$$u = (j + 0.5)s, \quad v = (i + 0.5)s, \quad (32)$$

where s is the image-to-feature downsampling factor, a depth bin d_k is unprojected as

$$x = \frac{(u - c_x)d_k}{f_x}, \quad y = \frac{(v - c_y)d_k}{f_y}, \quad z = d_k. \quad (33)$$

The corresponding voxel index in the OccuFly grid is

$$i_x = \left\lfloor \frac{x - x_{\min}}{\Delta_x} \right\rfloor, \quad i_y = \left\lfloor \frac{y - y_{\min}}{\Delta_y} \right\rfloor, \quad i_z = \left\lfloor \frac{z - z_{\min}}{\Delta_z} \right\rfloor. \quad (34)$$

Frustum features falling into the same voxel are aggregated by the FoundationSSC LSS module. The result is a low-resolution 3D feature volume that is further processed by proposal, VoxFormer-style reasoning [12], dual-feature fusion, and the FoundationSSC 3D decoder.

OccuFly geometry rewrite. FoundationSSC was originally configured for street-level SSC geometry. We therefore rewrite the geometry at module construction time before building LSS, proposal, VoxFormer, positional encoding, and Oc-cHead modules. With voxel downsampling [2, 2, 2], the low-resolution OccuFly grid is

$$(W_{\text{low}}, H_{\text{low}}, D_{\text{low}}) = (96, 64, 64), \quad (35)$$

corresponding to the full-resolution grid in Eq. (3). The rewritten configuration is

$$\text{point_cloud_range} = [-48, -32, 0, 48, 32, 64], \quad (36)$$

$$\text{xbound} = [-48, 48, 1.0], \quad \text{ybound} = [-32, 32, 1.0], \quad \text{zbound} = [0, 64, 1.0], \quad \text{dbound} = [0, 64, 1.0], \quad (37)$$

This rewrite prevents stale SemanticKITTI-like dimensions from entering the aerial model. The expected tensor shapes are

$$V^{\text{proposal}} \in \mathbb{R}^{B \times 1 \times 96 \times 64 \times 64}, \quad V^{\text{LSS}} \in \mathbb{R}^{B \times 128 \times 96 \times 64 \times 64}, \quad V^{\text{Vox}} \in \mathbb{R}^{B \times 128 \times 96 \times 64 \times 64}, \quad (38)$$

with final logits

$$O \in \mathbb{R}^{B \times |C| \times 128 \times 128 \times 192}. \quad (39)$$

The output is permuted into internal DHW order before loss and evaluation.

Depth detachment and warm starting. By default, the depth probability tensor passed to SSC is detached, and the V-JEPA backbone remains frozen. The depth frontend can be initialized from a monocular depth checkpoint, but the warm start loads only the depth metadata encoder, DPT decoder, and FiLM blocks. It does not load a separate DPT branch into the context adapter and does not update the V-JEPA encoder. This choice makes the SSC experiment a test of whether a fixed aerial depth prior can guide 3D semantics, rather than an unconstrained joint optimization in which SSC gradients may distort the depth estimator.

Pose-MVS-Cost-Volume FoundationSSC Hybrid The Pose-MVS-Cost-Volume FoundationSSC variant keeps the corrected OccuFly voxel geometry, LSS-style lifting interface, FoundationSSC hybrid view transformation, and SSC decoder from Sec. 3.6. It changes only the geometric frontend: instead of using a monocular DPT-FiLM depth-probability head, it constructs a metric depth-probability tensor from posed target/source video clips using the same reference-view cost-volume principle as MVSNet [29].

For each target frame, the dataset returns a target-centered context clip and a set of MVS view clips

$$X_t^{\text{ctx}} \in \mathbb{R}^{3 \times T \times H \times W}, \quad X_t^{\text{mvs}} = \{X_{t_i}\}_{i=0}^{N_s} \in \mathbb{R}^{(1+N_s) \times 3 \times T \times H \times W}, \quad t_0 = t, \quad (40)$$

where the first MVS view is the target and the remaining views are neighboring source frames. In the default configuration, $T = 4$ and $N_s = 2$, so the collated MVS tensor has shape $B \times 3 \times 3 \times 4 \times 384 \times 384$, where the second dimension is target plus two source views.

The context path and geometry path are intentionally source-decoupled. The context path encodes the target-centered clip, applies Dense Temporal Attention (DTA) within the target clip, and feeds the resulting DPT-compatible maps to a fresh DPT-FiLM dense feature head and the geometry-aware context adapter. The geometry path flattens the MVS view dimension, encodes each view clip

with the same frozen V-JEPA video backbone, applies DTA within each view, and reshapes the output back to view-specific feature maps:

$$G_i^{(l)} = E_\theta^{(l)}(X_{t_i}) \in \mathbb{R}^{C \times T' \times H_f \times W_f}, \quad F_i^{(l)} = A_l(G_i^{(l)}) \in \mathbb{R}^{C \times H_f \times W_f}, \quad (41)$$

where A_l is the DTA adapter for layer l , $H_f = W_f = 24$ for 384×384 inputs, and $T' = T/2$ for V-JEPA tubelets. Crucially, source views are not fused before matching; otherwise the geometry branch would lose the one-to-one association between a feature map and its camera pose.

A multi-layer view projector maps the selected V-JEPA layers to compact matching features. For each view i , the projected maps are resized to the first selected feature grid, averaged across layers, refined by shallow convolutions, and ℓ_2 -normalized:

$$\bar{F}_i = \text{norm} \left(\phi_{\text{mvs}}^{\text{proj}} \left(\frac{1}{|\mathcal{L}_G|} \sum_{l \in \mathcal{L}_G} \text{resize} \left(\pi_l(F_i^{(l)}) \right) \right) \right), \quad \bar{F}_i \in \mathbb{R}^{C_m \times H_f \times W_f}, \quad (42)$$

with $C_m = 64$ in the default configuration.

Let $T_i^{w \rightarrow c}$ be the world-to-camera pose for view i . The relative transform from target camera coordinates to source camera coordinates is

$$T_{i \leftarrow 0} = T_i^{w \rightarrow c} (T_0^{w \rightarrow c})^{-1} = \begin{bmatrix} R_{i \leftarrow 0} & t_{i \leftarrow 0} \\ 0 & 1 \end{bmatrix}. \quad (43)$$

Intrinsics are scaled from the resized image coordinates to the V-JEPA feature grid. For a target feature-grid coordinate $p = (u, v, 1)^\top$ and metric depth hypothesis d_k , the target-camera 3D point is

$$X_0(d_k, p) = d_k \tilde{K}_0^{-1} p, \quad (44)$$

and its projection into source view i is

$$p_i(d_k, p) \sim \tilde{K}_i (R_{i \leftarrow 0} X_0(d_k, p) + t_{i \leftarrow 0}). \quad (45)$$

The source feature map is bilinearly sampled at the normalized grid coordinate corresponding to $p_i(d_k, p)$. Invalid projections behind the source camera or outside the source feature grid are masked.

The raw projective cost volume is a valid-source average of cosine correlations, following the standard learned-MVS pattern of matching target/source features over metric depth hypotheses [29]:

$$S_{i,k}(p) = \langle \bar{F}_0(p), \text{sample}(\bar{F}_i, p_i(d_k, p)) \rangle, \quad (46)$$

$$C_k^{\text{MVS}}(p) = \frac{\sum_{i=1}^{N_s} V_{i,k}(p) S_{i,k}(p)}{\max \left(\sum_{i=1}^{N_s} V_{i,k}(p), 1 \right)}, \quad (47)$$

where $V_{i,k}(p)$ is the binary valid-warp mask. Depth planes without any valid source projection are assigned a fixed invalid cost. A lightweight 3D convolutional

regularizer, analogous to cost-volume regularization in MVSNet [29], converts the aggregated volume into raw depth logits,

$$Z^{\text{MVS}} = \rho_{3\text{D}}(C^{\text{MVS}}), \quad Z^{\text{MVS}} \in \mathbb{R}^{K \times H_f \times W_f}, \quad (48)$$

which are bilinearly upsampled to the FoundationSSC lifting resolution $H/8 \times W/8$.

The unconditioned Pose-MVS branch forms a categorical distribution directly from these logits:

$$P_{k,u,v}^{D,\text{MVS}} = \text{softmax}_k(Z_{k,u,v}^{\text{MVS}}), \quad \hat{D}_{u,v}^{\text{MVS}} = \sum_{k=1}^K P_{k,u,v}^{D,\text{MVS}} d_k. \quad (49)$$

With the default depth range and binning, $K = 256$ and the dense MVS depth map has shape $B \times 1 \times 48 \times 48$ for 384×384 input.

Pose-MVS-FiLM-Consistent FoundationSSC Hybrid The Pose-MVS-FiLM variant keeps the projective matching, source-view handling, cost-volume regularizer, FoundationSSC geometry, and SSC decoder unchanged, but conditions the MVS depth output using the same FiLM [18] metadata contract as the DPT-FiLM depth decoder. The metadata vector is

$$m_t = \left[\frac{h_t - 40}{10}, \frac{f_x}{W}, \frac{f_y}{H}, \frac{c_x}{W} - 0.5, \frac{c_y}{H} - 0.5 \right], \quad e_t = \eta(m_t) \in \mathbb{R}^{128}, \quad (50)$$

where η is the metadata encoder. A FiLM projection predicts depth-bin-wise affine parameters

$$(\gamma_t, \beta_t) = \omega(e_t), \quad \gamma_t, \beta_t \in \mathbb{R}^K, \quad (51)$$

and the projection ω is zero-initialized so that the FiLM path is the identity at initialization. The upsampled MVS logits are then conditioned before the depth softmax:

$$\tilde{Z}_{k,u,v}^{\text{MVS}} = (1 + \gamma_{t,k}) Z_{k,u,v}^{\text{MVS}} + \beta_{t,k}, \quad (52)$$

$$P_{k,u,v}^{D,\text{MVS-FiLM}} = \text{softmax}_k(\tilde{Z}_{k,u,v}^{\text{MVS}}). \quad (53)$$

The dense depth used downstream is not independently modulated. It is computed as the expectation under the same FiLM-conditioned distribution:

$$\hat{D}_{u,v}^{\text{MVS-FiLM}} = \sum_{k=1}^K P_{k,u,v}^{D,\text{MVS-FiLM}} d_k. \quad (54)$$

This consistency constraint is important because FoundationSSC consumes the categorical depth distribution for LSS-style lifting and the dense depth map for the voxel proposal/context pathway. If the two tensors encode different ray depths, the lifted feature volume and proposal prior can disagree. The

Table 2: SSC variants and the geometric prior isolated by each design.

Variant	Geometric prior	Question isolated
CGFormer comparison	Exported monocular priors	Do V-JEPA-derived depth priors improve a fixed CGFormer-style SSC evaluation path?
Monocular JEPA/F2D/FoundationSSC	V- DPT-FiLM depth probabilitySSC	Can single-image V-JEPA depth uncertainty guide FoundationSSC lifting for aerial SSC?
MDE warm start	DPT-FiLM prior initialized from a depth check-point	Does better monocular depth initialization improve downstream SSC?
Geometry-corrected FoundationSSC	Corrected OccuFly bounds and tensor shapes	Are gains caused by the intended model rather than stale street-scene geometry?
Semantic reweighting	Same corrected geometry with changed loss/class weights	Does rebalancing empty and occupied classes improve semantic mIoU?
Pose-MVS-Cost-Volume FoundationSSC	Projective multiview cost-volume probabilities	Does calibrated target/source correspondence provide a stronger 3D prior than monocular depth?
Pose-MVS-FiLM FoundationSSC	FiLM-conditioned MVS depth-bin probabilities	Does altitude/intrinsics conditioning improve the multiview geometric prior without changing the decoder?

Pose-MVS-FiLM branch therefore uses probability-first FiLM conditioning: altitude/intrinsics metadata modulates the depth-bin logits, the softmax gives the conditioned distribution, and the dense depth is exactly its metric expectation.

If the MVS depth-bin centers differ from the FoundationSSC LSS depth-bin centers, probability mass is rebinned in metric-depth space before LSS:

$$\bar{P}^{D,LSS} = \mathcal{R} \left(P^{D,MVS-FiLM}, \mathcal{B}_{MVS} \rightarrow \mathcal{B}_{LSS} \right), \quad \sum_k \bar{P}_{k,u,v}^{D,LSS} = 1. \quad (55)$$

The implementation preserves the predicted conditioned probability distribution when the source and target depth centers match, rather than regenerating a Gaussian distribution around the dense depth map.

The pair $(P_t^{D,MVS-FiLM}, \hat{D}_t^{MVS-FiLM})$ is passed to the same downstream FoundationSSC modules as the monocular and unconditioned MVS hybrids:

$$(C_t, P_t^{D,MVS-FiLM}, \hat{D}_t^{MVS-FiLM}, K_t) \rightarrow (V_t^{LSS}, V_t^{\text{proposal}}, V_t^{\text{Vox}}) \rightarrow \hat{Y}_t. \quad (56)$$

This isolates the effect of adding physically meaningful metadata conditioning to the multiview geometric prior while keeping the projective cost volume and voxel decoder fixed.

3.7 SSC Training Objective

The SSC objective combines semantic voxel classification, MonoScene-style scaling losses, optional occupancy BCE, and depth supervision. Let $O_q \in \mathbb{R}^{|C|}$ be the semantic logit vector at voxel q , let $p_{q,c} = \text{softmax}(O_q)_c$, and let $\Omega_V = \{q : Y_q \neq 255\}$. The semantic cross-entropy term is

$$\mathcal{L}_{\text{ssc-ce}} = -\frac{1}{|\Omega_V|} \sum_{q \in \Omega_V} w_{Y_q} \log p_{q,Y_q}, \quad (57)$$

where the empty/free-space weight is reduced to $w_0 = 0.25$ by default. Reweighting empty space is important in aerial occupancy grids because most voxels are empty, and unweighted CE can encourage high completion IoU at the expense of occupied semantic classes.

The semantic scaling loss encourages precision, recall, and specificity for each class rather than only per-voxel likelihood. For class c , define soft counts over valid voxels,

$$\text{TP}_c = \sum_{q \in \Omega_V} p_{q,c} \mathbf{1}\{Y_q = c\}, \quad (58)$$

$$\text{FP}_c = \sum_{q \in \Omega_V} p_{q,c} \mathbf{1}\{Y_q \neq c\}, \quad (59)$$

$$\text{FN}_c = \sum_{q \in \Omega_V} (1 - p_{q,c}) \mathbf{1}\{Y_q = c\}, \quad (60)$$

$$\text{TN}_c = \sum_{q \in \Omega_V} (1 - p_{q,c}) \mathbf{1}\{Y_q \neq c\}. \quad (61)$$

Precision, recall, and specificity are

$$\text{prec}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \epsilon}, \quad \text{rec}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c + \epsilon}, \quad \text{spec}_c = \frac{\text{TN}_c}{\text{TN}_c + \text{FP}_c + \epsilon}. \quad (62)$$

The semantic scaling loss applies binary cross-entropy toward 1 to these class-level quantities, averaged over classes present in the batch. The geometric scaling loss uses the same construction for binary occupancy, with target $o_q = \mathbf{1}\{Y_q \neq 0\}$ and predicted occupancy probability $p_q^{\text{occ}} = 1 - p_{q,0}$. This follows the motivation of prior monocular SSC losses: optimizing only voxel-wise CE can underemphasize small occupied structures and thin objects [5].

For the depth frontend inside SSC, we supervise both the probability distribution and the dense expectation. Given ground-truth depth D_i at valid pixel i , the soft target distribution is

$$Q_{i,k} = \frac{\exp\left(-\frac{(d_k - D_i)^2}{2\sigma_Q^2}\right)}{\sum_j \exp\left(-\frac{(d_j - D_i)^2}{2\sigma_Q^2}\right)}. \quad (63)$$

The depth classification loss is the soft cross-entropy

$$\mathcal{L}_{\text{depth-ce}} = -\frac{1}{|\Omega_D|} \sum_{i \in \Omega_D} \sum_k Q_{i,k} \log P_{i,k}^D, \quad (64)$$

which is equivalent to a KL divergence up to the entropy of Q . The relative depth L1 term is

$$\mathcal{L}_{\text{depth-l1}} = \frac{1}{|\Omega_D|} \sum_{i \in \Omega_D} \frac{|\hat{D}_i - D_i|}{\max(D_i, d_{\min})}. \quad (65)$$

For the pose-MVS branches, the same depth-supervision terms are applied to the cost-volume-derived distribution and expectation by substituting $P^D = P^{D,\text{MVS}}$ and $\hat{D} = \hat{D}^{\text{MVS}}$, or $P^D = P^{D,\text{MVS-FiLM}}$ and $\hat{D} = \hat{D}^{\text{MVS-FiLM}}$, in Eqs. (64) and (65). Thus the cost-volume projector, regularizer, and depth temporal adapters receive direct metric-depth supervision whenever the MVS depth branch is trainable. Freezing a randomly initialized MVS depth branch is not meaningful; frozen-depth ablations are reserved for the monocular DPT-depth branch.

The total optimized objective is

$$\mathcal{L} = \lambda_{\text{ce}} \mathcal{L}_{\text{ssc-ce}} + \lambda_{\text{sem}} \mathcal{L}_{\text{sem-scal}} + \lambda_{\text{geo}} \mathcal{L}_{\text{geo-scal}} + \lambda_{\text{occ}} \mathcal{L}_{\text{occ}} + \lambda_{\text{dce}} \mathcal{L}_{\text{depth-ce}} + \lambda_{\text{dl1}} \mathcal{L}_{\text{depth-l1}}. \quad (66)$$

Unless stated otherwise, we use

$$\lambda_{\text{ce}} = 1, \quad \lambda_{\text{sem}} = 1, \quad \lambda_{\text{geo}} = 1, \quad \lambda_{\text{occ}} = 0, \quad \lambda_{\text{dce}} = 0.5, \quad \lambda_{\text{dl1}} = 0.2. \quad (67)$$

The occupancy BCE term is still computed and logged when $\lambda_{\text{occ}} = 0$, but it does not contribute to the optimized total.

3.8 Evaluation Metrics

Depth models are evaluated only over valid pixels. We report root mean squared error (RMSE), absolute relative error (AbsRel), and log-RMSE:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i \in \Omega_D} (\hat{D}_i - D_i)^2}, \quad \text{AbsRel} = \frac{1}{N} \sum_{i \in \Omega_D} \frac{|\hat{D}_i - D_i|}{D_i}, \quad (68)$$

$$\text{LogRMSE} = \sqrt{\frac{1}{N} \sum_{i \in \Omega_D} (\log \hat{D}_i - \log D_i)^2}. \quad (69)$$

For video-depth runs we also compute the standard MAE, squared-relative error, and threshold accuracies $\delta_1, \delta_2, \delta_3$ when enabled by the evaluation script.

SSC evaluation follows the standard separation between scene completion and semantic scene completion. Completion metrics treat all non-empty labels

as occupied. If TP, FP, and FN are counted over occupied-vs-empty predictions, then

$$\text{IoU}_{\text{comp}} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN} + \epsilon}, \quad \text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP} + \epsilon}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN} + \epsilon}. \quad (70)$$

Semantic mIoU is computed over occupied classes only, excluding empty/free space and ignore label 255:

$$\text{mIoU} = \frac{1}{|\mathcal{C}_{\text{occ}}|} \sum_{c \in \mathcal{C}_{\text{occ}}} \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c + \epsilon}. \quad (71)$$

This metric choice is important for OccuFly because empty space dominates the voxel grid. Completion IoU measures whether the model predicts occupied volume at all, whereas occupied-class mIoU measures whether those occupied voxels receive meaningful aerial semantics. For the Pose-MVS and Pose-MVS-FiLM variants, depth metrics are computed on the target frame using the cost-volume expectation. We additionally log the valid-warp fraction and number of source views as diagnostics for whether the projected source evidence overlaps the target frustum. For Pose-MVS-FiLM, we also verify that $\sum_k P_{k,u,v}^{D,\text{MVS-FiLM}} = 1$ and that the dense depth equals the expectation in Eq. (54).

3.9 Implementation and Reproducibility Details

The backbone is frozen in all reported depth and SSC variants. Depth linear-probe experiments use distributed data parallel training with synchronized batch normalization and a global batch size of 8 in the verified launcher. The image DPT-FiLM and SFP variants train only decoder, depth-head, and FiLM parameters. The video DPT-FiLM+DTA variant trains the DTA adapters and DPT-FiLM decoder while keeping the video backbone fixed. Unless stated otherwise, depth heads are trained for 20 epochs with AdamW, weight decay, gradient clipping, and early stopping on the validation split; the DTA sweep uses the corresponding 30-epoch video configuration. These choices are deliberately lightweight relative to full backbone finetuning so that the experiments remain probes of V-JEPA 2.1 transfer.

The FoundationSSC hybrid uses the corrected OccuFly geometry rewrite described in Eq. (37). The geometry-correction run is therefore a configuration and tensor-shape correction, not a new feature-extraction architecture. The semantic-reweighting run changes loss balancing or class weighting over the corrected architecture. The MDE warm-start run initializes only the monocular depth frontend components described above and, by default, freezes them during SSC optimization. The Pose-MVS-Cost-Volume run is the architecture-level FoundationSSC variant that keeps the downstream FoundationSSC geometry and decoder fixed, but replaces the monocular depth-probability frontend with source-specific V-JEPA video features, pose-aware projective matching, a cost-volume regularizer, and cost-volume-derived depth probabilities. The Pose-MVS-FiLM run adds an identity-initialized metadata FiLM layer on the MVS depth-bin

logits and computes dense depth from the same conditioned probability distribution, so its gains should be attributed to metadata-conditioned multiview geometry rather than to a different voxel decoder. These distinctions are retained in the paper to avoid attributing gains to architectural changes that were in fact caused by geometry correction, loss balancing, initialization, or multiview correspondence.

4 Experimental Setup

4.1 Dataset and Tasks

We evaluate on OccuFly [8] for monocular metric depth estimation and semantic scene completion (SSC). For the standard protocol, we use scenes 01–05 for training, scenes 06–07 for validation, and scenes 08–09 for testing, across altitudes of 30, 40, and 50 m. The standard depth split contains 14,804 training frames, 1,965 validation frames, and 3,842 test frames. RGB inputs are resized to 384×384 . Depth values are represented in meters, clipped to $[0.001, 80]$, and evaluated only over finite positive pixels that remain valid after mask-aware resizing.

For SSC, OccuFly labels are defined on a voxel grid of shape $192 \times 128 \times 128$ in WHD order. The grid spans $x \in [-48, 48]$ m, $y \in [-32, 32]$ m, and $z \in [0, 64]$ m, corresponding to a voxel size of 0.5 m. Class 0 denotes empty/free space, and label 255 denotes invalid or unknown voxels ignored by the loss and metrics. FoundationSSC-style runs use the active SSC output taxonomy configured for the run; in the raw-ID-compatible OccuFly setting this corresponds to raw output slots $0, \dots, 36$, while CGFormer-style reporting remaps raw IDs to an empty-plus-occupied semantic taxonomy before computing semantic mIoU.

4.2 Depth Estimation Models

All V-JEPA-based depth experiments use a frozen V-JEPA2.1 [16] ViT-G/16 backbone unless otherwise stated. For 384×384 inputs, the patch size of 16 gives a 24×24 spatial patch grid. We evaluate five depth model families.

First, the linear-probe baseline applies a DINOv3-style [23] BatchNorm and 1×1 convolutional head to frozen V-JEPA features and predicts 256 depth-bin channels. Second, the image DPT-FiLM decoder uses frozen V-JEPA features from layers 11, 23, 37, and 47, a DPT [21] feature width of 256, and metadata conditioning through feature-wise linear modulation (FiLM) [18]. Third, the final-layer SFP DPT-FiLM variant replaces the four intermediate V-JEPA layers with a simple feature pyramid inspired by ViTDet [11], constructed from the final V-JEPA layer using scale factors 4, 2, 1, 0.5. Fourth, the video spatio-temporal linear probe (ST-LP) uses four-frame clips and preserves V-JEPA tubelet information by concatenating temporal tubelet features before a linear depth head. Finally, the video DPT-FiLM model uses four-frame clips and dense temporal attention over V-JEPA tubelet features at each selected layer.

4.3 Altitude Interpolation Protocol

In addition to the standard scene split, we evaluate an altitude interpolation stress test. This protocol trains on 30 m and 50 m samples from scenes 01–08, validates on 40 m samples from scenes 01–04, and tests on 40 m samples from scenes 05–08. The resulting split contains 11,493 training samples, 3,510 validation samples, and 2,648 test samples. This setting evaluates whether the model can interpolate to an unseen intermediate altitude rather than only generalize across held-out scenes.

4.4 Semantic Scene Completion Models

For SSC, we compare CGFormer-based baselines [31], monocular V-JEPA/FoundationSSC hybrids [6], a Pose-MVS-Cost-Volume FoundationSSC variant, and a Pose-MVS-FiLM FoundationSSC variant. The monocular hybrid architecture combines frozen V-JEPA image features, a DPT-FiLM depth and depth-probability head, semantic context features, Lift-Splat-Shoot-style frustum-to-voxel lifting [20], FoundationSSC-style 3D fusion, and an SSC decoder. This branch evaluates whether single-image V-JEPA depth uncertainty is sufficient to guide aerial semantic occupancy prediction.

The Pose-MVS-Cost-Volume FoundationSSC variant replaces the monocular depth-probability frontend with an explicit pose-aware multiview stereo cost-volume estimator [29]. For each target frame, the model uses a target-centered context clip and $N_s = 2$ posed source views by default. The source branch encodes target and source clips separately with the frozen V-JEPA video backbone, applies dense temporal attention within each view, preserves view identity, and constructs a projective depth-plane cost volume from the target/source features, resized intrinsics, and world-to-camera poses. The cost volume is regularized by a lightweight 3D CNN and converted into a categorical depth-probability tensor $P_t^{D,MVS} \in \mathbb{R}^{256 \times 48 \times 48}$ and dense depth expectation $\hat{D}_t^{MVS} \in \mathbb{R}^{1 \times 48 \times 48}$. These outputs are passed to the same FoundationSSC lifting, proposal, 3D fusion, and SSC decoder used by the monocular hybrid. This design isolates the effect of calibrated multiview correspondence while keeping the downstream voxel geometry and decoder fixed.

The Pose-MVS-FiLM variant keeps the same projective cost-volume geometry but adds altitude/intrinsics FiLM conditioning [18] directly to the MVS depth output. It encodes the metadata vector $[h_{\text{norm}}, f_x/W, f_y/H, c_x/W - 0.5, c_y/H - 0.5]$ with the same metadata contract used by DPT-FiLM, then applies FiLM to the upsampled MVS depth-bin logits before the softmax over depth. The dense depth supplied to FoundationSSC is computed as the expectation of this same FiLM-conditioned probability distribution, ensuring that the probability volume used by LSS and the metric depth map used by voxel proposal remain geometrically consistent. The FiLM affine projection is zero-initialized, so the initial model is equivalent to the unconditioned Pose-MVS branch and can learn altitude- or calibration-dependent corrections during SSC training.

We evaluate ablations covering depth-prior choice, monocular-depth-estimation warm-starting, FoundationSSC geometry correction, semantic loss reweighting, Pose-MVS cost-volume depth estimation, and Pose-MVS-FiLM output conditioning.

4.5 Training Details

Depth models are trained with AdamW and a masked scale-invariant depth loss unless otherwise specified. The linear probe is trained for 20 epochs with learning rate 1.5×10^{-3} , weight decay 10^{-3} , gradient clipping at 35, batch size 1 per GPU, and 8-GPU distributed data parallel training. The ST-LP model is trained for 20 epochs with learning rate 3×10^{-4} , weight decay 10^{-4} , batch size 1 per GPU, and 8 GPUs. The image DPT-FiLM and SFP DPT-FiLM models are trained for 20 epochs with gradient clipping at 35. The video DPT-FiLM model is trained for 30 epochs with eight temporal-attention heads.

SSC hybrids use AdamW parameter groups for the depth, context-DPT, context, view-transformation, and decoder modules. Unless otherwise stated, SSC training uses cosine learning-rate scheduling with one warmup epoch, and checkpoints are selected using validation SSC mIoU, occupied IoU, and depth RMSE. For the Pose-MVS and Pose-MVS-FiLM variants, the V-JEPA video backbone is frozen, while the MVS cost-volume head, depth metadata encoder, depth temporal adapters, context temporal adapters, context DPT-FiLM head, context adapter, view-transformation modules, and SSC decoder are trainable unless disabled by an ablation. The default Pose-MVS-FiLM configuration uses video input, four-frame clips, real temporal context, two source views, world-to-camera poses, 64 MVS matching channels, 16 cost-volume regularizer channels, 256 depth bins, target downsampling by 8, batch size 1 per GPU, 20 epochs, depth learning rate 10^{-4} , depth weight decay 10^{-4} , and learning rate 3×10^{-4} for context-DPT, context, view, and decoder groups.

All experiments are run on an NVIDIA DGX node with 8 NVIDIA H100 80GB GPUs. We omit private hostnames, usernames, absolute paths, and local checkpoint directories for double-blind review.

4.6 Evaluation Metrics

For depth estimation, we report RMSE, AbsRel, and log-RMSE over valid pixels. For video depth models, we additionally report δ_1 , δ_2 , and δ_3 threshold accuracies. For SSC, we report occupied IoU, semantic mIoU over occupied semantic classes, class-wise IoU, and altitude-stratified metrics where applicable. For Pose-MVS variants, depth metrics are computed on the target frame from the cost-volume depth expectation, and SSC metrics are computed on the target-frame voxel grid using the same protocol as the monocular SSC models. We also log MVS diagnostic quantities such as valid-warp fraction and source-view count to identify pose, intrinsics, or overlap failures. For the Pose-MVS-FiLM variant, we additionally verify that the FiLM-conditioned depth probability is

Table 3: Font sizes of headings. Table captions should always be positioned *above* the tables.

Heading level	Example	Font size and style
Title (centered)	Lecture Notes . . .	14 point, bold
1st-level heading	1 Introduction	12 point, bold
2nd-level heading	2.1 Printing Area	10 point, bold
3rd-level heading	Headings. Text follows . . .	10 point, bold
4th-level heading	<i>Remark.</i> Text follows . . .	10 point, italic

normalized over depth and that the dense depth equals the expectation under the conditioned probability distribution.

5 Results and Discussion

5.1 Headings

Headings should be capitalized (*i.e.*, nouns, verbs, and all other words except articles, prepositions, and conjunctions should be set with an initial capital) and should, with the exception of the title, be aligned to the left. Only the first two levels of section headings should be numbered, as shown in Tab. 3. The respective font sizes are also given in Tab. 3. Kindly refrain from using “0” when numbering your section headings. Words joined by a hyphen are subject to a special rule. If the first word can stand alone, the second word should be capitalized.

Here are some examples of headings: “Criteria to Disprove Context-Freeness of Collage Languages”, “On Correcting the Intrusion of Tracing Non-deterministic Programs by Software”, “A User-Friendly and Extendable Data Distribution System”, “Multi-flip Networks: Parallelizing GenSAT”, “Self-determinations of Man”.

5.2 Figures

For L^AT_EX users, we recommend integrating figures in your paper using the package `graphicx`.

It is essential that all illustrations are clear and legible. Vector graphics (rather than rasterized images) should be used for diagrams and schemas whenever possible. Please check that the lines in line drawings are not interrupted and have a constant width. Line drawings are to have a resolution of at least 800 dpi (preferably 1200 dpi). Grids and details within figures must be clearly legible and may not be written one on top of the other. The lettering in figures should not use font sizes smaller than 6 pt (~2 mm character height).

Figures should be numbered and should have a caption, which should always be positioned *under* the figures, in contrast to the caption belonging to a table,

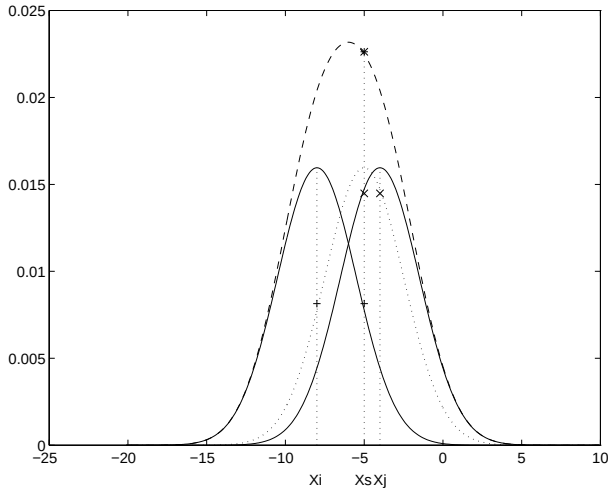


Fig. 1: One kernel at x_s (*dotted kernel*) or two kernels at x_i and x_j (*left and right*) lead to the same summed estimate at x_s . This shows a figure consisting of different types of lines. Elements of the figure described in the caption should be set in italics, in parentheses, as shown in this sample caption. The last sentence of a figure caption should generally end with a full stop, except when the caption is not a full sentence.



(a) An example of a subfigure



(b) Another example of a subfigure

Fig. 2: Centered, short example caption

which should always appear *above* the table. Figures and Tables should be cross-referred in the text.

If they are short, they are centered between the margins (*cf.* Fig. 2). Longer captions, covering more than one line, are justified (Fig. 1 shows an example). Captions that do not constitute a full sentence, do not have a period.

If possible (*e.g.*, if you use $\text{\LaTeX}$) please define figures as floating objects. $\text{\LaTeX}$ users, please avoid using the location parameter “h” for “here”. If you have to insert a pagebreak before a figure, please ensure that the previous page is completely filled.

5.3 Formulas

Displayed equations or formulas are centered and set on a separate line (with an extra line or half line space above and below). Equations should be numbered for reference. The numbers should be consecutive within the contribution, with

numbers enclosed in parentheses and set on the right margin. For example,

$$\psi(u) = \int_0^T \left[\frac{1}{2} (A_0^{-1}u, u) + N^*(-u) \right] dt \quad (72)$$

$$= 0 \quad (73)$$

and

$$E = m \cdot c^2. \quad (74)$$

Please do not include section counters in the numbering.

Numbering equations makes reviewing more efficient, because reviewers can refer to a line on a page. It is important for readers to be able to refer to any particular equation. Just because you did not refer to it in the text does not mean some future reader might not need to refer to it. It is cumbersome to have to use circumlocutions like “the equation second from the top of page 3”. (Note that the ruler will not be present in the final copy, so is not an alternative to equation numbers). All authors will benefit from reading Mermin’s description of how to write mathematics: <https://doi.org/10.1063/1.2811173>.

Equations should never be in color and should be punctuated in the same way as ordinary text. They should not be pasted in as figures.

Lemmas, Propositions, and Theorems. The numbers accorded to lemmas, propositions, and theorems, *etc.* should appear in consecutive order, starting with Lemma 1. Please do not include section counters in the numbering like “Theorem 1.1”.

5.4 Footnotes.

The superscript numeral used to refer to a footnote appears in the text either directly after the word to be discussed or – in relation to a phrase or a sentence – following the punctuation mark (comma, semicolon, or period).¹ For remarks pertaining to the title or the authors’ names, in the header of a paper, symbols should be used instead of a number. Please note that no footnotes may be included in the abstract.

5.5 Cross References

For the benefit of author(s) and readers, please use the

`\cref{...}`

command for cross-referencing to figures, tables, equations, or sections. This will automatically insert the appropriate label alongside the cross reference as in this example:

¹ The footnote numeral is set flush left and the text follows with the usual word spacing. Second and subsequent lines are indented.

To see how our method outperforms previous work, please see Fig. 1 and Tab. 3. It is also possible to refer to multiple targets as once, *e.g.* to Figs. 1 and 2a. You may also return to ?? or look at Eq. (74).

If you do not wish to abbreviate the label, for example at the beginning of the sentence, you can use the

`\Cref{...}`

command. Here is an example:

Figure 1 is also quite important.

5.6 Program Code

Program listings or program commands in the text are normally set in typewriter font (*e.g.*, `printf("Hello world!\n");`).

5.7 Citations

Arabic numbers are used for citation, which is sequential either by order of citation or by alphabetical order of the references, depending on which sequence is used in the list of references. The reference numbers are given in brackets and are not superscript. Please observe the following guidelines:

- Single citation: [?]
- Multiple citation: [?, ?, ?, ?]. The numbers should be listed in numerical order. If you use the template as advised, this will be taken care of automatically.
- If an author’s name is used in the text: Alpher [?] was the first ...

Please write all references using the Latin alphabet. If the title of the book you are referring to is, *e.g.*, in Russian or Chinese, then please write (in Russian) or (in Chinese) at the end of the transcript or translation of the title. All references cited in the text should be in the list of references and vice versa.

References should be formatted with the official LNCS reference style. The L^AT_EX template already takes care of that through the use of the `splncs04.bst` BibT_EX style file. Springer strongly encourages you to include DOIs (Digital Object Identifiers) in your references (*cf.* [?]). The DOI is a unique code allotted by the publisher to each online paper or journal article. It provides a stable way of finding published papers and their metadata. The insertion of DOIs increases the overall length of the references section, but this should not concern you as the reference section is not counted toward the page limit.

5.8 Miscellaneous

Compare the following:

<code>\$conf_a\$</code>	$conf_a$
<code>\$\mathit{conf}_a\$</code>	$conf_a$

See The $\text{\TeX}$ book, p. 165.

The space after *e.g.*, meaning “for example”, should not be a sentence-ending space. So *e.g.* is correct, *e.g.* is not. The provided `\eg` macro takes care of this.

When citing a multi-author paper, you may save space by using “et alia”, shortened to “*et al.*” (not “*et. al.*” as “*et*” is a complete word). If you use the `\etal` macro provided, then you need not worry about double periods when used at the end of a sentence as in Alpher *et al.* However, use it only when there are three or more authors. Thus, the following is correct: “Frobnication has been trendy lately. It was introduced by Alpher [?], and subsequently developed by Alpher and Fotheringham-Smythe [?], and Alpher *et al.* [?].”

This is incorrect: “... subsequently developed by Alpher *et al.* [?] ...” because reference [?] has just two authors.

6 Conclusion

The paper ends with a conclusion.

6.1 Camera-Ready Manuscript Preparation

This information will follow after paper decisions have been announced.

Page 33 of the manuscript.

Page 35 of the manuscript.

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Page 36 of the manuscript. This is the last page.

Now we have reached the maximum length of an ACCV 2026 submission (excluding references). References should start immediately after the main text, but can continue past p. 14 if needed.

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